



Sydney Girls High School

**2019**

YEAR 11

# Chemistry

|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|

Student Number

Booklet 1 of 2

Part A & B

## General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- Two pages of data sheets and a Periodic Table are provided
- Write your student number on the multiple choice answer sheet and on each booklet

## Total marks – 35

### Part A – Pages 2-5

#### 10 marks

- Attempt Questions 1 – 10

### Part B – Pages 6-8

#### 13 marks

- Attempt Questions 11 – 14

### Part C – Pages 9 – 11

#### 12 marks

- Attempt Questions 15 – 16

**Part A – 10 marks****Attempt Questions 1 – 10**

Use the multiple-choice answer sheet for Questions 1–10

---

Question 1 and 2 refer to the following information.

A group of students conducted an experiment to investigate the reactivity of four different transition metals (A, B, C and D). They placed small pieces of each metal in separate test tubes with the appropriate test solutions. The result are summarised in the table below.

| Metal ions<br>Metal | $A^{3+}(aq)$    | $B^{2+}(aq)$    | $C^{4+}(aq)$ | $D^{3+}(aq)$    |
|---------------------|-----------------|-----------------|--------------|-----------------|
| A                   |                 | No reaction     | No reaction  | Reaction occurs |
| B                   | Reaction occurs |                 | No reaction  | Reaction occurs |
| C                   | Reaction occurs | Reaction occurs |              | Reaction occurs |
| D                   | No reaction     | No reaction     | No reaction  |                 |

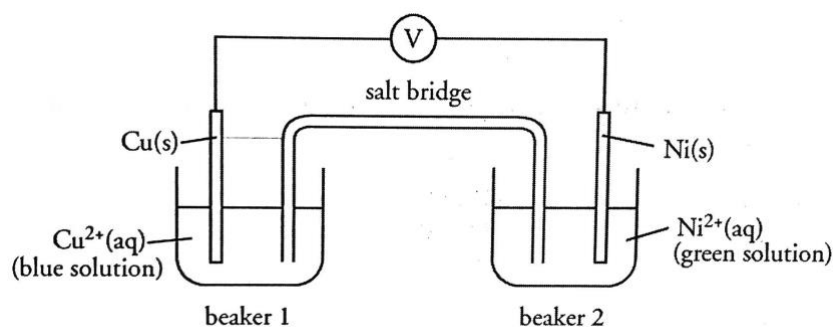
1 Which of these metals is the most easily oxidised?

- (A) Metal A
- (B) Metal B
- (C) Metal C
- (D) Metal D

2 Which of these metals is the weakest reducing agent?

- (A) Metal A
- (B) Metal B
- (C) Metal C
- (D) Metal D

- 3 Which half equation best represents the change when sodium metal reacts with dilute hydrochloric acid?
- (A)  $\text{Na}^{2+} + 2\text{e}^- \rightarrow \text{Na(s)}$
- (B)  $\text{Na}^{2+} \rightarrow \text{Na(s)} + 2\text{e}^-$
- (C)  $\text{Na(s)} \rightarrow \text{Na}^+ + \text{e}^-$
- (D)  $\text{Na(s)} + \text{e}^- \rightarrow \text{Na}^+$
- 4 The diagram below shows an electrochemical cell which has copper and nickel electrodes.



What three observations for this cell are correct?

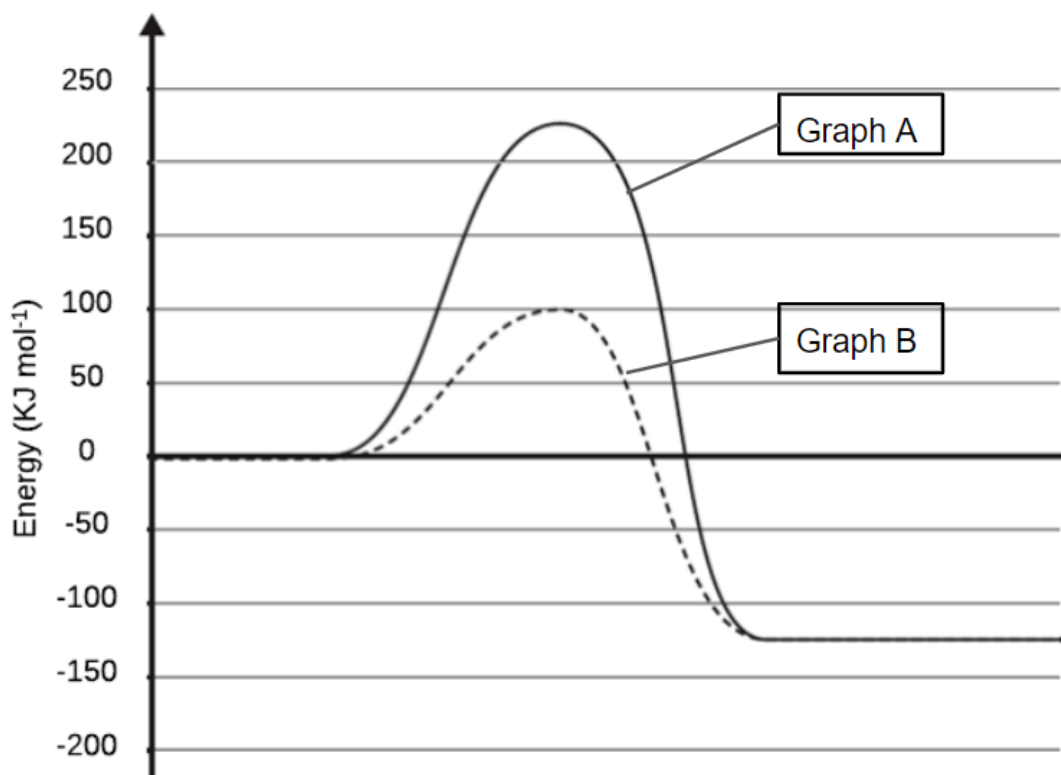
|     | Observation 1                                   | Observation 2                                   | Observation 3                                    |
|-----|-------------------------------------------------|-------------------------------------------------|--------------------------------------------------|
| (A) | The reading shown on the voltmeter was -0.57 V. | The solution in beaker 1 became a darker blue.  | The solution in beaker 2 became a fainter green. |
| (B) | The reading shown on the voltmeter was -0.11 V. | A red/brown solid form on the copper electrode. | The solution in beaker 1 became a fainter blue.  |
| (C) | A grey solid formed on the nickel electrode.    | In beaker 1 the solution became a darker blue.  | The solution in beaker 2 became a darker green.  |
| (D) | The nickel electrode was the negative terminal. | A red/brown solid form on the copper electrode. | The solution in beaker 2 became a darker green.  |

- 5 The transition metal manganese exists in a range of different oxidation states.
- The oxidation states of Mn in  $\text{MnO}_2$  and  $\text{MnO}_4^-$  respectively are,

- (A) +2 and +7
- (B) +3 and +6
- (C) +4 and +7
- (D) +6 and +4

Question 6 and 7 refer to the following information.

The energy profile diagram of a particular reaction is shown below. Graph A represent the uncatalysed reaction and Graph B represent the catalysed reaction.



6 Which of the following is correct?

|     | $E_a$ of catalysed reaction (kJ mol <sup>-1</sup> ) | $\Delta H$ of uncatalysed reaction (kJ mol <sup>-1</sup> ) |
|-----|-----------------------------------------------------|------------------------------------------------------------|
| (A) | 100                                                 | -125                                                       |
| (B) | 100                                                 | -350                                                       |
| (C) | 225                                                 | -125                                                       |
| (D) | 225                                                 | -350                                                       |

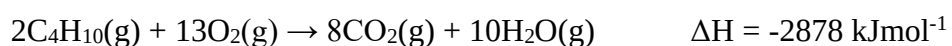
7 What would be the activation energy of the reverse of the reaction in Graph A?

- (A) 100 kJ mol<sup>-1</sup>
- (B) 225 kJ mol<sup>-1</sup>
- (C) 250 kJ mol<sup>-1</sup>
- (D) 350 kJ mol<sup>-1</sup>

- 8 When hydrochloric acid reacts with potassium hydroxide, energy is released because
- (A) the reaction involves the breaking of bonds.
  - (B) the stored energy in the products is greater than the stored energy in the reactants.
  - (C) the activation energy is greater than the energy released from the hydrogen and oxygen molecules.
  - (D) as the bonds in the products form, more energy is released than is required to break the bonds in the reactants.
- 9 Under what circumstances will a change be nonspontaneous regardless of the temperature?

|     | $\Delta H$ | $\Delta S$ | $-T\Delta S$ | $\Delta G$ |
|-----|------------|------------|--------------|------------|
| (A) | +          | –          | +            | +          |
| (B) | –          | +          | –            | –          |
| (C) | –          | –          | +            | +          |
| (D) | +          | +          | –            | –          |

- 10 The equation for the reaction of butane with oxygen is



When 22g of carbon dioxide is produced in this reaction, the energy change will be:

- (A) 180 kJ released
- (B) 360 kJ released
- (C) 1439 kJ released
- (D) 2878 kJ released

|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|

Student Number

## Part B – 13 marks

### Attempt Questions 11 - 14

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

#### Question 11 (4 marks)

The formulae of 10 substances are given below:

|                               |                |                                |                   |                 |
|-------------------------------|----------------|--------------------------------|-------------------|-----------------|
| NaOH                          | KCl            | AgNO <sub>3</sub>              | CaCO <sub>3</sub> | N <sub>2</sub>  |
| C <sub>2</sub> H <sub>6</sub> | O <sub>2</sub> | H <sub>2</sub> SO <sub>4</sub> | H <sub>2</sub>    | CO <sub>2</sub> |

- a) Identify TWO substances that would react together in an aqueous environment to form a precipitate:

- i) Write a balanced overall chemical equation including states. 1K

.....

- ii) Write a net ionic equation including states. 1K

.....

- b) Which TWO substances will take part in an acid and base reaction in an aqueous environment?

- i) Write a balanced overall chemical equation including states. 1K

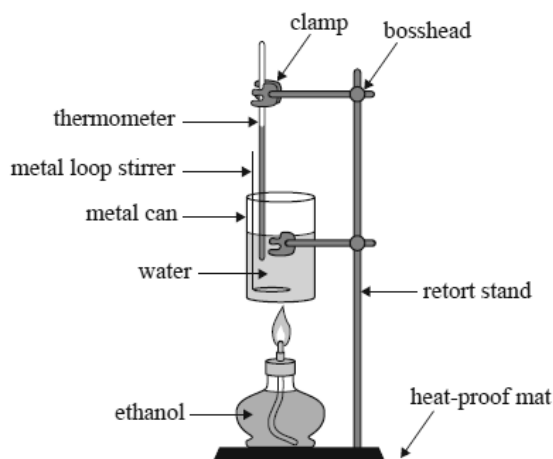
.....

- ii) Write a net ionic equation including states. 1K

.....

### Question 12 (4 marks)

Students were asked to perform a first-hand investigation to determine the molar heat of combustion of ethanol ( $\text{C}_2\text{H}_5\text{OH}$ ). The following extract is from the practical report of one student.



Lab data:

|                              |           |
|------------------------------|-----------|
| Mass of water                | = 150.0 g |
| Initial mass of burner       | = 182.5 g |
| Final mass of burner         | = 181.1 g |
| Initial temperature of water | = 24.0°C  |
| Final temperature of water   | = 64.0°C  |

After completing the calculations correctly, the student found that the answer did not agree with the value found on data sheets.

- a) Suggest TWO reasons for this. 2P

.....

.....

.....

.....

- b) Propose TWO ways the student can improve the accuracy of the results. 2P

.....

.....

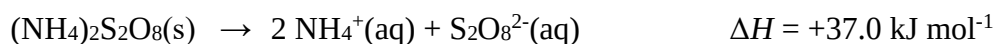
.....

.....

.....

**Question 13 (3 marks)**

The compound ammonium peroxydisulfate,  $(\text{NH}_4)_2\text{S}_2\text{O}_8$ , dissolves in water according to the following equation.



In an experiment, 22.82 g of ammonium peroxydisulfate was dissolved in 100 g of water. If the final temperature of the water was  $15.0^\circ\text{C}$ , what is the initial temperature? 3A

.....

.....

.....

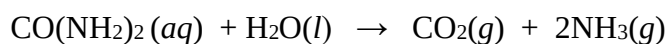
.....

.....

.....

**Question 14 (2 marks)**

The following equation shows the reaction of urea with water at  $25^\circ\text{C}$ ,



The enthalpy change  $\Delta H^\circ$  for this reaction is  $+119.2 \text{ kJ mol}^{-1}$  and the entropy change  $\Delta S^\circ$  is  $+354.8 \text{ J mol}^{-1} \text{ K}^{-1}$ . Calculate the Gibbs free energy change, then predict the spontaneity of this reaction. Show all working. 2A

.....

.....

.....

.....

.....

**END OF PART B**

|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|

Student Number

**Part C – 12 marks****Attempt Questions 15 - 16**

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

---

**Question 15 (10 marks)**

- a) You conducted an experiment to determine the effect of concentration on rate of reaction. Describe a method to safely investigate this, including a relevant chemical equation.

5P

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

**Question 15 - continued**

- b) With the aid of a diagram, explain the effect of increasing the concentration of reactants on the rate of reaction. 3K

.....

.....

.....

.....

.....

- c) Explain the effect of increasing temperature on the rate of reaction. 2K

.....

.....

.....

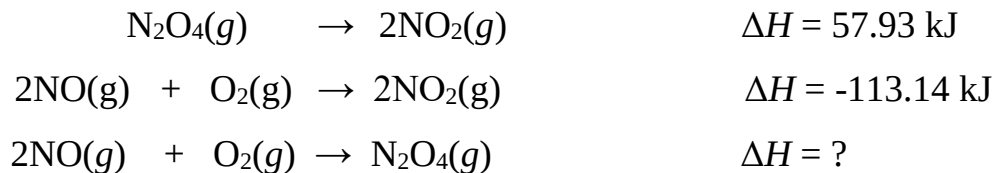
.....

.....

.....

**Question 16 (2 marks)**

Use the following equations to calculate the enthalpy of dinitrogen tetroxide formation. 2A  
Show all working.



.....  
.....  
.....  
.....  
.....

**END OF PART C**



Sydney Girls High School

**2019**

YEAR 11

# Chemistry

|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|

Student Number

Booklet 1 of 2

Part A & B

## General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- Two pages of data sheets and a Periodic Table are provided
- Write your student number on the multiple choice answer sheet and on each booklet

## Total marks – 35

### Part A – Pages 2-5

#### 10 marks

- Attempt Questions 1 – 10

### Part B – Pages 6-8

#### 13 marks

- Attempt Questions 11 – 14

### Part C – Pages 9 – 11

#### 12 marks

- Attempt Questions 15 – 16

**Part A – 10 marks****Attempt Questions 1 – 10**

Use the multiple-choice answer sheet for Questions 1–10

---

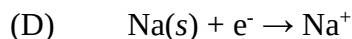
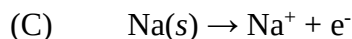
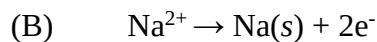
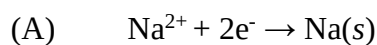
Question 1 and 2 refer to the following information.

A group of students conducted an experiment to investigate the reactivity of four different transition metals (A, B, C and D). They placed small pieces of each metal in separate test tubes with the appropriate test solutions. The result are summarised in the table below.

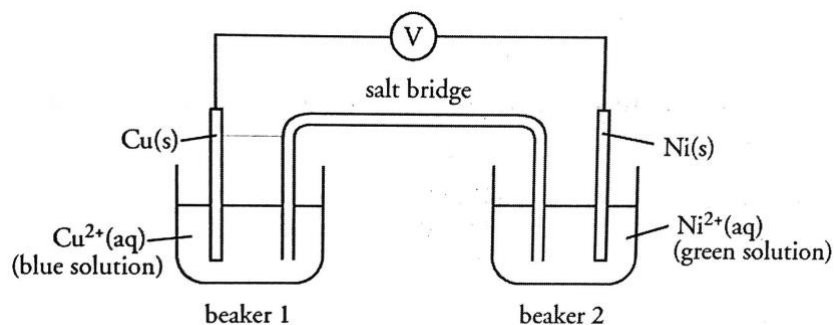
| Metal ions<br>Metal | $A^{3+}(aq)$    | $B^{2+}(aq)$    | $C^{4+}(aq)$ | $D^{3+}(aq)$    |
|---------------------|-----------------|-----------------|--------------|-----------------|
| A                   |                 | No reaction     | No reaction  | Reaction occurs |
| B                   | Reaction occurs |                 | No reaction  | Reaction occurs |
| C                   | Reaction occurs | Reaction occurs |              | Reaction occurs |
| D                   | No reaction     | No reaction     | No reaction  |                 |

- Which of these metals is the most easily oxidised?  
  
(A) Metal A  
(B) Metal B  
(C) Metal C  
(D) Metal D
- Which of these metals is the weakest reducing agent?  
  
(A) Metal A  
(B) Metal B  
(C) Metal C  
(D) Metal D

3 Which half equation best represents the change when sodium metal reacts with dilute hydrochloric acid?



4 The diagram below shows an electrochemical cell which has copper and nickel electrodes.

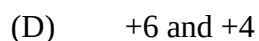
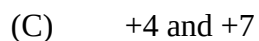
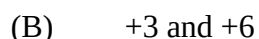
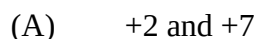


What three observations for this cell are correct?

|     | Observation 1                                   | Observation 2                                   | Observation 3                                    |
|-----|-------------------------------------------------|-------------------------------------------------|--------------------------------------------------|
| (A) | The reading shown on the voltmeter was -0.57 V. | The solution in beaker 1 became a darker blue.  | The solution in beaker 2 became a fainter green. |
| (B) | The reading shown on the voltmeter was -0.11 V. | A red/brown solid form on the copper electrode. | The solution in beaker 1 became a fainter blue.  |
| (C) | A grey solid formed on the nickel electrode.    | In beaker 1 the solution became a darker blue.  | The solution in beaker 2 became a darker green.  |
| (D) | The nickel electrode was the negative terminal. | A red/brown solid form on the copper electrode. | The solution in beaker 2 became a darker green.  |

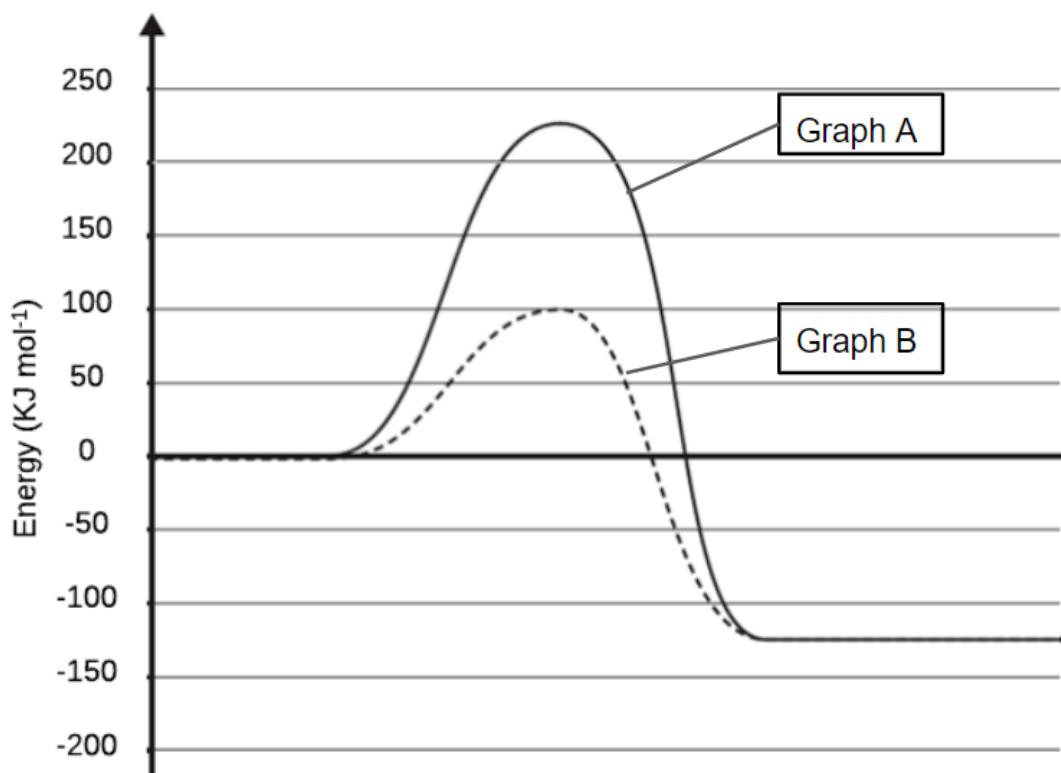
5 The transition metal manganese exists in a range of different oxidation states.

The oxidation states of Mn in  $\text{MnO}_2$  and  $\text{MnO}_4^-$  respectively are,



Question 6 and 7 refer to the following information.

The energy profile diagram of a particular reaction is shown below. Graph A represent the uncatalysed reaction and Graph B represent the catalysed reaction.



6 Which of the following is correct?

|     | $E_a$ of catalysed reaction (kJ mol <sup>-1</sup> ) | $\Delta H$ of uncatalysed reaction (kJ mol <sup>-1</sup> ) |
|-----|-----------------------------------------------------|------------------------------------------------------------|
| (A) | 100                                                 | -125                                                       |
| (B) | 100                                                 | -350                                                       |
| (C) | 225                                                 | -125                                                       |
| (D) | 225                                                 | -350                                                       |

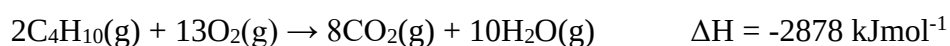
7 What would be the activation energy of the reverse of the reaction in Graph A?

- (A) 100 kJ mol<sup>-1</sup>
- (B) 225 kJ mol<sup>-1</sup>
- (C) 250 kJ mol<sup>-1</sup>
- (D) 350 kJ mol<sup>-1</sup>

- 8 When hydrochloric acid reacts with potassium hydroxide, energy is released because
- (A) the reaction involves the breaking of bonds.
- (B) the stored energy in the products is greater than the stored energy in the reactants.
- (C) the activation energy is greater than the energy released from the hydrogen and oxygen molecules.
- (D) as the bonds in the products form, more energy is released than is required to break the bonds in the reactants.
- 9 Under what circumstances will a change be nonspontaneous regardless of the temperature?

|     | $\Delta H$ | $\Delta S$ | $-T\Delta S$ | $\Delta G$ |
|-----|------------|------------|--------------|------------|
| (A) | +          | –          | +            | +          |
| (B) | –          | +          | –            | –          |
| (C) | –          | –          | +            | +          |
| (D) | +          | +          | –            | –          |

- 10 The equation for the reaction of butane with oxygen is



When 22g of carbon dioxide is produced in this reaction, the energy change will be:

- (A) 180 kJ released
- (B) 360 kJ released
- (C) 1439 kJ released
- (D) 2878 kJ released

|   |   |    |   |
|---|---|----|---|
| 1 | C | 2  | D |
| 3 | C | 4  | D |
| 5 | C | 6  | A |
| 7 | D | 8  | D |
| 9 | A | 10 | A |

|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|

Student Number

## Part B – 13 marks

### Attempt Questions 11 - 14

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

#### Question 11 (4 marks)

The formulae of 10 substances are given below:

|                               |                |                                |                   |                 |
|-------------------------------|----------------|--------------------------------|-------------------|-----------------|
| NaOH                          | KCl            | AgNO <sub>3</sub>              | CaCO <sub>3</sub> | N <sub>2</sub>  |
| C <sub>2</sub> H <sub>6</sub> | O <sub>2</sub> | H <sub>2</sub> SO <sub>4</sub> | H <sub>2</sub>    | CO <sub>2</sub> |

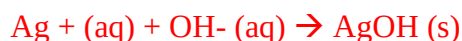
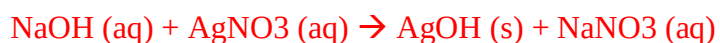
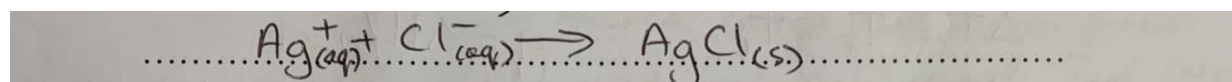
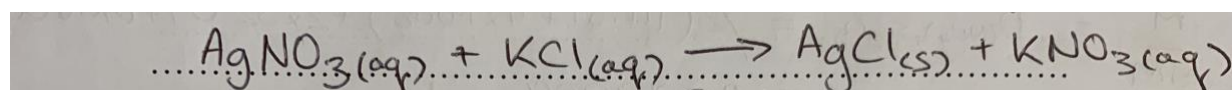
- a) Identify TWO substances that would react together in an aqueous environment to form a precipitate:

i) Write a balanced overall chemical equation including states. 1K

ii) Write a net ionic equation including states.

1K

|                                                        |   |
|--------------------------------------------------------|---|
| Correct balanced chemical equation with correct states | 1 |
| Correct net ionic equation with correct states         | 1 |



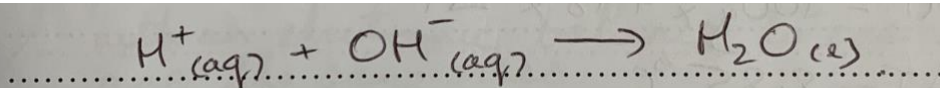
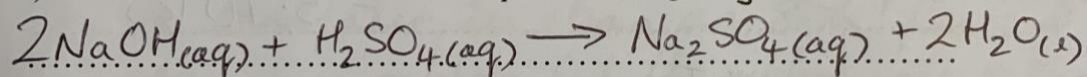
- calcium carbonate has very low solubility 0.013 g/L at 25°C

b) Which TWO substances will take part in an acid and base reaction in an aqueous environment?

i) Write a balanced overall chemical equation including states. 1K

ii) Write a net ionic equation including states. 1K

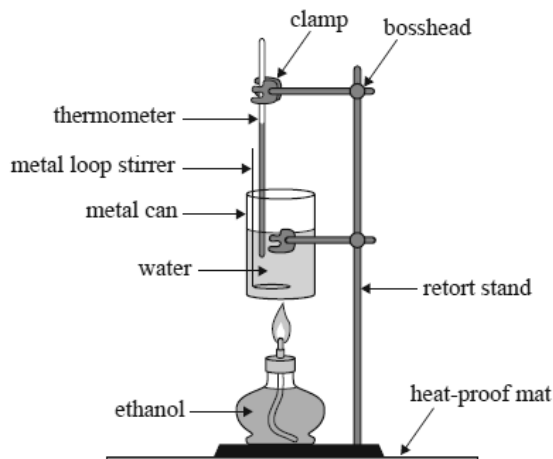
|                                                        |   |
|--------------------------------------------------------|---|
| Correct balanced chemical equation with correct states | 1 |
| Correct net ionic equation with correct states         | 1 |



- Simplify  $2\text{H}^+_{(aq)} + \text{OH}^-_{(aq)} \rightarrow \text{H}_2\text{O}_{(l)}$

### Question 12 (4 marks)

Students were asked to perform a first-hand investigation to determine the molar heat of combustion of ethanol ( $\text{C}_2\text{H}_5\text{OH}$ ). The following extract is from the practical report of one student.



Lab data:

|                              |           |
|------------------------------|-----------|
| Mass of water                | = 150.0 g |
| Initial mass of burner       | = 182.5 g |
| Final mass of burner         | = 181.1 g |
| Initial temperature of water | = 24.0°C  |
| Final temperature of water   | = 64.0°C  |

After completing the calculations correctly, the student found that the answer did not agree with the value found on data sheets.

a) Suggest TWO reasons for this.

2P

.....

|                                                                                                                                                                                                                |          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Two correct reasons provided</b> <ul style="list-style-type: none"><li>• Heat loss to surroundings</li><li>• Standard conditions (298K and 1 atm) might not apply</li><li>• Incomplete combustion</li></ul> | <b>2</b> |
| <b>One Correct reason provided</b>                                                                                                                                                                             | <b>1</b> |

b) Propose TWO ways the student can improve the accuracy of the results.

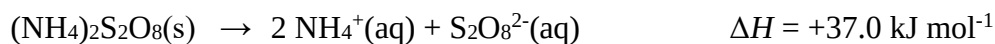
2P

|                                                                                                                                                                                                                                                                                                                                                |          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Two correct improvements provided</b><br>e.g. <ul style="list-style-type: none"><li>• keep the distance between the wick and the calorimeter as low as possible</li><li>• a lid or insulation could be used to reduce heat loss of the calorimeter</li><li>• ensure the thermometer is not touching the bottom of the calorimeter</li></ul> | <b>2</b> |
| <b>One Correct improvement provided</b>                                                                                                                                                                                                                                                                                                        | <b>1</b> |

- Using more precise equipment will not improve the reasons provided in a) significantly

**Question 13 (3 marks)**

The compound ammonium peroxydisulfate,  $(\text{NH}_4)_2\text{S}_2\text{O}_8$ , dissolves in water according to the following equation.



In an experiment, 22.82 g of ammonium peroxydisulfate was dissolved in 100 g of water. If the final temperature of the water was  $15.0^\circ\text{C}$ , what is the initial temperature? 3A

|                                                                                                     |   |
|-----------------------------------------------------------------------------------------------------|---|
| • The number of mole for ammonium peroxydisulfate is correctly calculated $n = 0.09999 \text{ mol}$ | 1 |
| • Correctly find $q = \Delta H \times n = 3.70 \text{ kJ}$                                          | 1 |
| • Correct substitution into $q = mc\Delta T$ and find $T = 23.9^\circ\text{C}$                      | 1 |

$$\frac{m}{MM} = \frac{22.82}{284.04} = 0.09999 \text{ mol}$$

$$\Delta T = T_f - T_i \quad q = \Delta H \times n$$

$$= 37.0 \text{ kJ mol}^{-1} \times 0.1 \text{ mol}$$

$$= 3.70 \text{ kJ}$$

$$q = mc\Delta T$$

$$3.70 = 100 \text{ g} \times 4.18 \times \Delta T$$

$$\Delta T = -8.85^\circ\text{C}$$

$$T_i = T_f - \Delta T$$

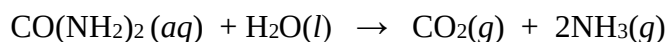
$$= 15 - 8.85$$

$$= 23.85$$

$$= 24^\circ\text{C} \text{ (to 2 sig figs)}$$

**Question 14 (2 marks)**

The following equation shows the reaction of urea with water at 25 °C,



The enthalpy change  $\Delta H^\circ$  for this reaction is +119.2 kJ mol<sup>-1</sup> and the entropy change  $\Delta S^\circ$  is +354.8 J mol<sup>-1</sup> K<sup>-1</sup>. Calculate the Gibbs free energy change, then predict the spontaneity of this reaction. Show all working.

2A

|                                                                                                                 |   |
|-----------------------------------------------------------------------------------------------------------------|---|
| • Correct substitution into $\Delta G = \Delta H - T\Delta S$ , and find $\Delta G = +13.5 \text{ kJ mol}^{-1}$ | 1 |
| • Correct prediction from the sign of $\Delta G$                                                                | 1 |

$$\begin{aligned}\Delta G &= \Delta H - T\Delta S \\ &= +119.2 \text{ kJ mol}^{-1} - 298 \times 354.8 \text{ J mol}^{-1} \text{ K}^{-1} \\ &= +119.2 \times 10^3 \quad 105730 \\ &= 119200 \text{ J mol}^{-1} - 105730 \text{ J mol}^{-1} \\ &= +13.5 \text{ kJ mol}^{-1}.\end{aligned}$$

As  $\Delta G$  is +ve, reaction will not be spontaneous at this temperature.

END OF PART B

Booklet 2 of 2

|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|

Student Number

## Part C – 12 marks

### Attempt Questions 15 - 16

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

#### Question 15 (10 marks)

- a) You conducted an experiment to determine the effect of concentration on rate of reaction. Describe a method to safely investigate this, including a relevant chemical equation.

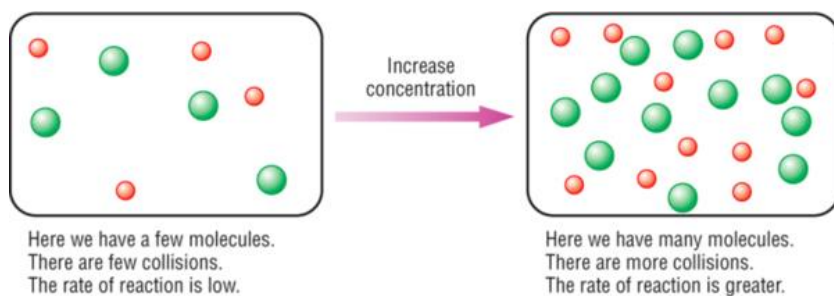
5P

|                                                                                                                                                                                                                                                                                                                                                                        |     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <ul style="list-style-type: none"> <li>Consider the safety aspect of the experiment</li> <li>Experiment must be fair, controlled, observable and repeatable</li> <li>Correct independent and dependent variables identified</li> <li>All equipment used must be identified (stopwatch, electronic balance)</li> <li>One relevant chemical equation included</li> </ul> | 5   |
| Missing one or more points                                                                                                                                                                                                                                                                                                                                             | 2-4 |
| Some relevant information                                                                                                                                                                                                                                                                                                                                              | 1   |

- b) With the aid of a diagram, explain the effect of increasing the concentration of reactants on the rate of reaction.

3K

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <ul style="list-style-type: none"> <li>Increasing concentration of reactants increases rate of reaction</li> <li>Increasing the concentration means there are more reactant particle/ increases the frequency of collisions between reactants in a given space (volume). This increases the chance that reactant particles will collide. The increased frequency of collisions results in a faster rate of reaction.</li> <li>Relevant labelled diagram provided</li> </ul> | 3 |
| Missing one point as indicate above                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2 |
| One correct point provided                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1 |



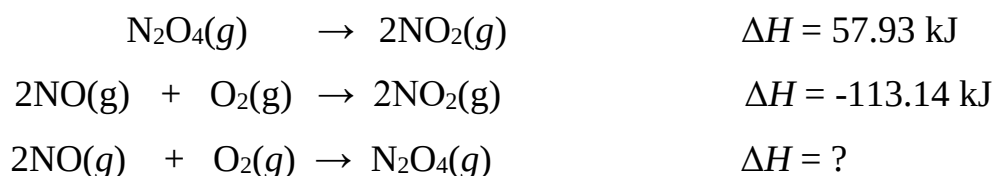
c) Explain the effect of increasing temperature on the rate of reaction.

2K

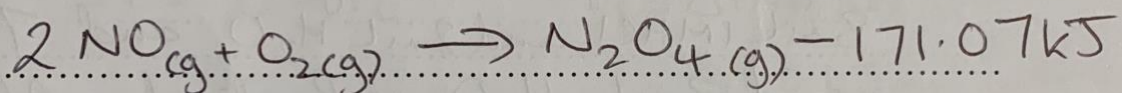
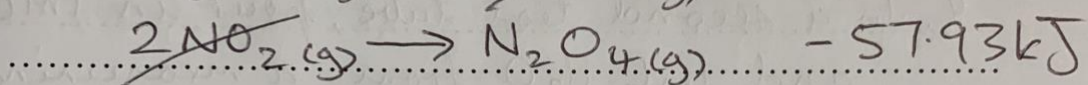
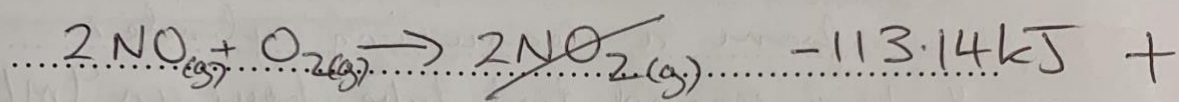
|                                                                                                                                                                                                                                                                                                                                                         |   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <ul style="list-style-type: none"> <li>Increasing temperature increases rate of reaction</li> <li>As the temperature increases 1. the particles move faster resulting in more collisions which in turn leads to a faster reaction. And 2. There are more collisions which have enough activation energy to result in a successful collision.</li> </ul> | 2 |
| One correct reason provided                                                                                                                                                                                                                                                                                                                             | 1 |

### Question 16 (2 marks)

Use the following equations to calculate the enthalpy of dinitrogen tetroxide formation. 2A  
Show all working.



|                                                                                                                                                                                            |   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <ul style="list-style-type: none"> <li>Correctly rearrange the first equation<br/><math>2\text{NO}_2(g) \rightarrow \text{N}_2\text{O}_4(g) \Delta H = -57.93 \text{ kJ}</math></li> </ul> | 1 |
| <ul style="list-style-type: none"> <li>Correctly adding the equation together and find <math>\Delta H = -171.07 \text{ kJ}</math>, with the correct unit</li> </ul>                        | 1 |
| * Lose 1 mark if there is no correct unit provided                                                                                                                                         |   |



**END OF PART C**